



Source: Wikimedia

The Curiosity Mars rover is perhaps the best example of a robot that is maximally autonomous - it takes round about half an hour to receive a signal (which may be a distress call!) from Curiosity, process it, and send back appropriate instructions

ming and corner-case testing, and the implementation of at least some degree of machine learning, data processing and artificial intelligence.

- **Bottom-Up (the brain-last approach):** This is a somewhat more ad hoc view of things. One attempts to associate every end-effector with its own sensor, thus identifying an input for every control output, or 'behaviour'. These behaviours, which are nothing but sensor-effector pairs, must then be arranged in a certain order of priority. This priority order can be called a hierarchy. In general, such hard-coding is not going to produce a very efficient or 'intelligent' robot. However, specific sets of parameters can be found that produce very good performance. This, coupled with the ability to dynamically adjust the priority order in realtime, make this method an acceptable ad hoc solution in most cases.

How it learns - the Artificial Intelligence revolution

With the surge in availability of data, sensors, and huge streams of information, a revolution in 'data science' has been born. It poses the question of how to decompose huge volumes of unintelligible information into useful insights or sets of parameters.